

# Interconnecting Series and Parallel ODEs

## Local equations, connection constraints and phasors

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2026-10-08

### Abstract

ODE blocks connect by sharing one terminal variable and summing the other. A mixed series/parallel connection yields four state equations, a scalar fourth-order ODE, and exact phasor formulas. Reconstruction preserves the internal coordinates and their initial values.

PDF created: 2026-10-08 13:07:40 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-ode-interconnect-ser-par>

## 1 Local equations and connection rules

Let  $D = d/dt$ . Each block has terminal voltage  $e_j$  and current  $i_j$ . Orient series voltage drops along the common current, and parallel currents in the same direction between common terminals. All  $L, R, C$  parameters are positive constants.

The same algebra applies wherever the ODEs and connection balances correspond. Translate variables, coefficients and units with [the ODE template](#), Sections 2–3. Its force–voltage map sends electrical series to mechanical parallel, and electrical parallel to mechanical series.

A series LRC block  $\mathcal{S}$  uses charge  $q$ :

$$e_s = L_s D^2 q + R_s Dq + \frac{q}{C_s}, \quad i_s = Dq. \quad (1)$$

A parallel CRL block  $\mathcal{P}$  uses flux linkage  $\lambda$ :

$$i_p = C_p D^2 \lambda + \frac{D\lambda}{R_p} + \frac{\lambda}{L_p}, \quad e_p = D\lambda. \quad (2)$$

Either block can enter either connection. Series imposes

$$\boxed{i_1 = i_2 = i, \quad e = e_1 + e_2,} \quad (3)$$

and parallel imposes

$$\boxed{e_1 = e_2 = e, \quad i = i_1 + i_2.} \quad (4)$$

For more blocks, extend the sums and retain each internal coordinate. These orientations also fix the signs for the terminal-power accounts in [Energy Ledgers for Forced Harmonic ODEs](#), Sections 1–2.

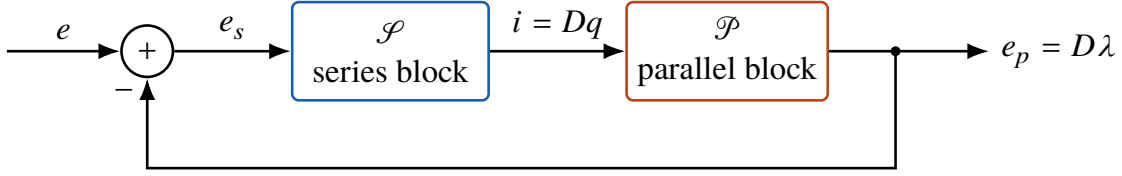
## 2 A mixed interconnection

Connect  $\mathcal{S}$  and  $\mathcal{P}$  in series. With total voltage  $e$ , common current  $i$ , and internal voltage  $e_p$ , substitute  $e_s = e - e_p$  and  $i_s = i_p = i$  into the local equations:

$$L_s D^2 q + R_s Dq + \frac{q}{C_s} + D\lambda = e, \quad (5)$$

$$C_p D^2 \lambda + \frac{D\lambda}{R_p} + \frac{\lambda}{L_p} = Dq. \quad (6)$$

Each block supplies the other's terminal driving quantity:  $D\lambda = e_p$  and  $Dq = i$ .



**Figure 1.** Schematic of the mixed connection. The series block receives  $e - e_p$  and determines  $i$ ; the parallel block receives  $i$  and determines  $e_p$ .

Keeping those terminal variables gives four first-order equations:

$$\begin{aligned} Dq &= i, \\ L_s Di &= e - R_s i - \frac{q}{C_s} - e_p, \\ C_p De_p &= i - \frac{e_p}{R_p} - \frac{\lambda}{L_p}, \\ D\lambda &= e_p. \end{aligned} \quad (7)$$

The state and initial data are

$$x = (q, i, e_p, \lambda)^\top, \quad x(t_0) = (q_0, i_0, e_{p,0}, \lambda_0)^\top. \quad (8)$$

Connecting the terminals couples the two ODEs while retaining all four initial coordinates.

## 3 The connection in phasor form

For  $x(t) = \text{Re}(\hat{x}e^{i\omega t})$ ,  $i^2 = -1$  and  $\omega > 0$ , substitute  $D = i\omega$ :

$$\begin{aligned} Z_s &= i\omega L_s + R_s + \frac{1}{i\omega C_s}, & \hat{e}_s &= Z_s \hat{i}, \\ Y_p &= i\omega C_p + \frac{1}{R_p} + \frac{1}{i\omega L_p}, & \hat{i} &= Y_p \hat{e}_p. \end{aligned} \quad (9)$$

Here  $Z_s$  converts current to voltage and  $Y_p$  converts voltage to current. The series constraints remain

$$\hat{e} = Z_s \hat{i} + \hat{e}_p, \quad \hat{i} = Y_p \hat{e}_p. \quad (10)$$

Writing  $\Delta = 1 + Z_s Y_p$  gives

$$\boxed{\hat{e}_p = \frac{\hat{e}}{\Delta}, \quad \hat{i} = \frac{Y_p \hat{e}}{\Delta}, \quad \Delta \neq 0.} \quad (11)$$

Recover  $\hat{q} = \hat{i}/(i\omega)$  and  $\hat{\lambda} = \hat{e}_p/(i\omega)$ . The parallel currents are

$$\hat{i}_C = i\omega C_p \hat{e}_p, \quad \hat{i}_R = \frac{\hat{e}_p}{R_p}, \quad \hat{i}_L = \frac{\hat{e}_p}{i\omega L_p}, \quad \hat{i}_C + \hat{i}_R + \hat{i}_L = \hat{i}. \quad (12)$$

These are harmonic components. The full solution still needs the initial state in (8).

### 3.1 Further blocks

Series voltages give  $Z_{eq} = \sum_j Z_j$ ; parallel currents give  $Y_{eq} = \sum_j Y_j$ . Thus

$$\hat{e}_j = \frac{Z_j}{\sum_\ell Z_\ell} \hat{e} \quad (\text{series}), \quad \hat{i}_j = \frac{Y_j}{\sum_\ell Y_\ell} \hat{i} \quad (\text{parallel}). \quad (13)$$

Use these quotients only with nonzero denominators. Where finite and nonzero,  $Y = 1/Z$  lets a nested series/parallel group be combined from its inner branches outward. Sum complex phasors before taking magnitudes; at a zero denominator, check compatibility in the original linear equations.

## 4 Elimination without losing initial data

Assume  $e(t)$  is twice continuously differentiable and define

$$A(s) = L_s s^2 + R_s s + \frac{1}{C_s}, \quad B(s) = C_p s^2 + \frac{s}{R_p} + \frac{1}{L_p}. \quad (14)$$

The coupled equations are  $A(D)q + D\lambda = e$  and  $B(D)\lambda = Dq$ . Apply  $B(D)$  to the first and use constant coefficients to commute it with  $D$ :

$$\boxed{K(D)q = B(D)e, \quad K(s) = A(s)B(s) + s^2.} \quad (15)$$

This fourth-order equation contains both local polynomials and the  $s^2$  term from connecting their terminal derivatives.

Reconstruct the eliminated variables by

$$i = Dq, \quad e_p = e - A(D)q, \quad \lambda = L_p \left( i - C_p D e_p - \frac{e_p}{R_p} \right). \quad (16)$$

The scalar equation implies  $B(D)e_p = D^2 q$ , so differentiating the last formula gives  $D\lambda = e_p$ . All original equations are recovered.

To recover the original initial state, set  $q(t_0) = q_0$ ,  $(Dq)(t_0) = i_0$  and

$$\begin{aligned} (D^2 q)(t_0) &= \frac{e(t_0) - R_s i_0 - q_0/C_s - e_{p,0}}{L_s}, \\ (D^3 q)(t_0) &= \frac{(De)(t_0) - R_s (D^2 q)(t_0) - i_0/C_s - (De_p)(t_0)}{L_s}, \end{aligned} \quad (17)$$

where  $(De_p)(t_0) = (i_0 - e_{p,0}/R_p - \lambda_0/L_p)/C_p$ . The four scalar initial values encode the same four internal coordinates.

Since  $Z_s = A(i\omega)/(i\omega)$  and  $Y_p = B(i\omega)/(i\omega)$ ,

$$K(i\omega) = (i\omega)^2 \Delta. \quad (18)$$

Hence  $\hat{q} = B(i\omega)\hat{e}/K(i\omega)$  and reconstruction give (11) wherever the denominator is nonzero.

To extend a suitable local equation, choose higher-order coefficients from *ODE Coefficient Synthesis*, Sections 1–4, and reuse the terminal constraints. Once the local equations are known, this assembles the system without repeating mesh analysis for the full network. Phasors still use  $D = i\omega$ ; added differential orders require corresponding initial data consistent with the connection.

## References

1. H. Nilre and B. C. Herlin (2026). [The ODE Template and Its Domain Equivalents](#).
2. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
3. H. Nilre and B. C. Herlin (2026). [Energy Ledgers for Forced Harmonic ODEs](#).